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Experiment to determine rate of photosynthesis by Wilmott's bubbler

Wilmott's bubbler consists of a wide mouth bottle fitted with single holed cork, a glass tube with lower end having wider opening to insert *Hydrilla* plant, the upper end fitted to a narrow bottle with water (Figure 13.25).

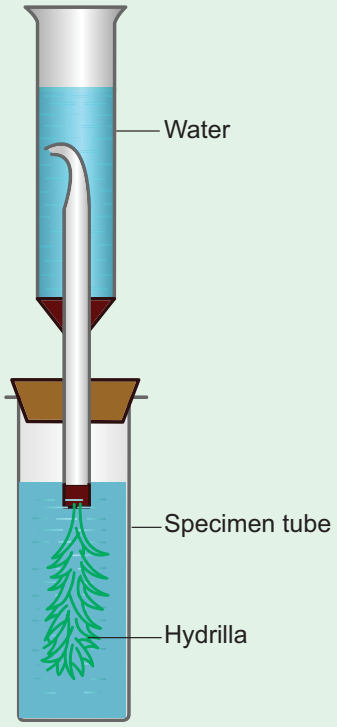


Figure 13.25: Wilmott's Bubbler

1. Fill the bottle with water and insert *Hydrilla* twig into the wider part of the tube
2. *Hydrilla* plant should be cut inside the water to avoid entry of air bubbles
3. Fix the tube with jar which acts as water reservoir
4. Keep the apparatus in sunlight
5. Count the bubbles when they are in same size.

3. Accumulation of Carbohydrates

Photosynthetic end products like carbohydrates are accumulated in cells and if translocation of carbohydrates is slow then this will affect the rate of photosynthesis.

4. Anatomy of leaf

Thickness of cuticle and epidermis, distribution of stomata, presence or absence of Kranz anatomy and relative proportion of photosynthetic cells affect photosynthesis.

5. Hormones

Hormones like gibberellins and cytokinin increase the rate of photosynthesis.

Test tube funnel experiment or Experiment to prove oxygen evolved during Photosynthesis

1. Place *Hydrilla* plant at the bottom of a beaker containing water.
2. Cover the plant with an inverted funnel.
3. Invert a test tube over the funnel.
4. Keep this setup in sunlight.

Note your observations (Figure 13. 26).

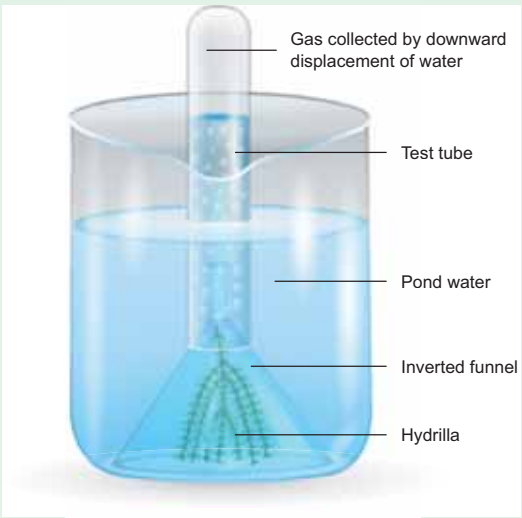


Figure 13.26: Test tube funnel experiment